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Using EPECs to Model Bilevel Games in Restructured Electricity Markets with Locational Prices

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We study a bilevel noncooperative game-theoretic model of restructured electricity markets, with locational marginal prices. Each player in this game faces a bilevel optimization problem that we model as a mathematical program with equilibrium constraints (MPEC). The corresponding game is an example of an equilibrium program with equilibrium constraints (EPEC). We establish sufficient conditions for the existence of pure-strategy Nash equilibria for this class of bilevel games and give some applications. We show by examples the effect of network transmission limits, i.e., congestion, on the existence of equilibria. Then we study, for more general equilibrium programs with equilibrium constraints, the weaker pure-strategy concepts of local Nash and Nash stationary equilibria. We pose the latter as solutions of complementarity problems (CPs) and show their equivalence with the former in some cases. Finally, we present numerical examples of methods that attempt to find local Nash equilibria or Nash stationary points of randomly generated electricity market games.

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1. Introduction

Game-theoretic models are employed to investigate strategic behavior in restructured or deregulated electricity markets—see Berry et al. (1999), Cardell et al. (1997), Hobbs et al. (2000), Nasser (1998), Oren (1997), Seeley et al. (2000), and Stoft (1999) to mention just a few references—and, of course, more general markets. Such models depend on the structure and rules of the market, and, for electricity and other distributed markets, the way network constraints are handled. Using the bilevel electricity market game of Berry et al. (1999) as a jumping-off point, this paper seeks general modeling paradigms of bilevel games, and in doing so provides a working knowledge of equilibrium programs with equilibrium constraints (EPECs), from models to methods.

Our investigation has two aspects. First, in §3 in the context of Berry et al. (1999), we seek to understand when pure-strategy Nash equilibria exist and when they may not. Second and more generally (§§4–5), by modeling each player's problem as a mathematical program with equilibrium constraints (MPEC), we recast bilevel games as EPECs. Stationarity theory for MPECs allows us to introduce Nash stationary points, a weakening of the Nash equilibrium concept, for EPECs, and show that the format of a standard mixed complementarity problem (CP) captures such points. This opens the way to new analysis and algorithms.

A particular motivation is the area of noncooperative games that model electricity markets over a network of generators and consumers in the style of Berry et al. (1999); see also Backerman et al. (2000), Cardell et al. (1997), Hobbs et al. (2000), and Hu et al. (2004) for a brief review of these and related market models. A coordinator or central operator, called an independent system operator (ISO), schedules the dispatch quantities, prices, and transmission of electricity. To do so, the ISO solves an optimization problem that relies on the configuration and electrical properties of the network, and on bids from generators and consumers (e.g., retailers) that propose how price should vary with quantity. This results in a game that has a bilevel structure; in particular, the ISO's optimization problem, which lies on the lower level, gives rise to nondifferentiability and nonquasiconcavity of an individual participant's profit function on the upper level. This is the setting of the first aspect of our study.

The second aspect of our study is broader, namely, games where each player faces a bilevel maximization problem

that can be modeled as an MPEC. Such problems have nonconvex constraints, and therefore may have multiple local maxima. Thus, the corresponding game, an EPEC, may not have any (pure-strategy) Nash equilibria in some cases, e.g., Example 4 in §3.3. See also Berry et al. (1999, Footnote 8, p. 143) and Weber and Overbye (1999). Thus, we study alternatives to the Nash equilibrium concept, aimed at model formats that are suitable for the application of powerful mathematical programming solvers. Our computational approach is to seek points such that each player's strategy satisfies the stationary conditions for his or her MPEC, and to apply this in electricity market models.

Although the emphasis of this paper is on existence, uniqueness, and algorithms for Nash equilibria, a companion paper (Hu et al. 2004) applies the methods of this paper to investigate the economic issues, such as market power, effects of congestion on the bidding behavior, comparison of different market structures, etc. Both papers are derived from the first author's PhD dissertation (Hu 2003). A preliminary version of this paper was presented in Hu and Ralph (2001). See also Berry et al. (1999) for discussion of economic implications of outcomes of these electricity market models.

An alternative CP approach to EPECs in electricity markets is given in Wang et al. (2004). Leyffer and Munson (2005) give various complementarity and MPEC reformulations of EPECs in multileader multifollower games. See also the PhD dissertation by Ehrenmann (2004a); various applications in economics such as Ehrenmann (2004b), Ehrenmann and Neuhoff (2003), Hu et al. (2004), Murphy and Smeers (2002), Su (2004a), and Yao et al. (2007); and the algorithm investigation in Su (2004b).

We mention in passing that without a Nash equilibrium, the questions "Is the model deficient?" and "Is the market unstable?" point to future investigations that might study the gap between models and their application in policy, commerce, and regulation. This is well beyond our scope here.

This paper is organized as follows. Section 2 briefly reviews the formulation of the ISO's problem and the associated related bilevel game of Berry et al. (1999). Section 3 is mainly a study of sufficient conditions for existence of pure-strategy Nash equilibria for the game scenario in which only the linear part of the cost or utility function is bid—the so-called bid-a-only scenario. It also gives examples showing how transmission limits may (may not) affect existence of Nash equilibria. In this section, the network and associated pricing and dispatch problems are implicit, which simplifies the format of the game at the expense of requiring analysis of nonsmooth payoff functions. Section 4 studies more general bilevel games, which are reformulated as EPECs. It proposes local Nash equilibria and Nash stationary equilibria as weaker alternatives to Nash equilibria, and develops a CP formulation of Nash stationary points to which standard software can be applied. Finally, returning to electricity markets, it is shown that Nash stationary points are actually local Nash equilibria for the bid-a-only game scenario in some circumstances. Section 5 presents numerical examples of two approaches to finding equilibria. The first approach is to apply the solver PATH (Dirkse and Ferris 1995, Ferris and Munson 1999) to the CP formulation of Nash stationary conditions, and the second approach uses a kind of fixed-point iteration, called *diagonalization*, that is implemented using the nonlinear programming solver CONOPT (Drud 1994). PATH is found to be particularly robust in identifying local Nash equilibria, whereas diagonalization holds some promise for efficiency on larger problems. Section 6 concludes the paper.

2. Bilevel Game Formulation

2.1. Pricing and Dispatch by the Optimal Power Flow Model

As many others do, we work with a lossless direct current network, which consists of nodes where generators and consumers are located, and (electricity) lines or links connecting nodes; see Chao and Peck (1997). For simplicity of notation, assume that there are N nodes with one generator at each node, $i = 1, \ldots, s$, and one consumer at each node, $i = s+1, \ldots, N$. The ordering of players does not play any role other than simplifying notation and does not restrict us from multiple players of the same type at any node. However, we do not model an interesting situation where one generator may own several generating units at different nodes. For further simplicity, only two types of network constraints (power balance and line flow constraints) are included in the ISO's social cost minimization problem.

The market works as follows. Generators and consumers bid cost and utility functions, respectively, to the ISO. These are revealed to all parties but may differ from the actual functions. The ISO determines the dispatch quantities of generation and consumption by minimizing the social cost, the difference between the total of cost bids and the total of utility bids, subject to network constraints. (Equivalently, the ISO could maximize social welfare.) The price at a node is set to the marginal value of the associated cost or utility bid, reflecting the fact that higher prices may be needed to curb demand at network nodes to which transmission constraints limit the delivery of electricity. For further development of the locational or nodal pricing idea, see Chao and Peck (1997) and Hogan (1992).

In practical markets, each generator typically bids a piecewise-linear cost function (or its supply bid is a step function). We use convex quadratic cost and negative utility function bids, as in Berry et al. (1999), Cardell et al. (1997), and Weber and Overbye (1999). This allows rough approximation of piecewise-linear functions, and importantly for later analysis, ensures that the ISO's problem has a unique solution. A generator at node i bids $c_i(q_i) = a_i q_i + b_i q_i^2$ with $q_i \geqslant 0$, where q_i is the quantity of generation and a_i, b_i are bid parameters, and a consumer at node j bids $-c_j(q_j) = -a_j q_j - b_j q_j^2$ with $q_j \leqslant 0$. Let $Q = Q(b) = \operatorname{diag}(b_1, \ldots, b_N)$, $a = (a_1, \ldots, a_N)^T$, $q = (q_1, \ldots, q_N)^T$.

Let $\mathscr{L}$ be the set of all links. A link l from node i to node j has a transmission limit (capacity) denoted C_l or C_{ij}. The ISO's optimal power flow problem can be written, e.g., Wu and Varaiya (1999), as

$$\begin{aligned} \underset{q}{\text{minimize}} \quad & q^T Q q + a^T q \\ \text{subject to} \quad & q_1 + \cdots + q_N = 0, \\ & -C_{\max} \leqslant \Phi q \leqslant C_{\max}, \\ & q_i \geqslant 0, \quad i = 1, \ldots, s, \\ & q_i \leqslant 0, \quad i = s+1, \ldots, N, \end{aligned} \tag{1}$$

where $C_{\max}$ is the vector of link capacities; Φ is a $|\mathscr{L}| \times N$ matrix of distribution factors (Wood and Wollenberg 1996) or injection shift factors (Liu and Gross 2004); and $|\mathscr{L}|$ is the number of transmission lines in the network. Distribution factors are determined by three sets of parameters: an arbitrary but fixed reference node or swing bus, the topology of the network (connections between nodes by lines), and electrical properties of lines such as susceptance. In particular, $\Phi_{ij,k}$ describes the change in the flow on line i–j that results from injecting one more unit of power at node k and extracting this at the reference node. It is worth pointing out that the solution of (1) is independent of the reference node.

Observe that the optimal power flow problem is a quadratic program with a nonempty feasible set and has a strictly convex objective if each $b_i > 0$. The next result is well known, e.g., Luo et al. (1996, pp. 169–171) or the more general results surveyed in Bonnans and Shapiro (1998, §5):

Lemma 1. *Let $C_{\max}$ be a positive vector.*

(i) *Fix $b > 0$, and consider a as a parameter vector in* (1). *Then, problem* (1) *is solvable for any a, its optimal solution $q^* = q^*(a)$ is unique, and $q^*(a)$ is a piecewise-linear mapping in a.*

(ii) *Consider a and b as parameters in* (1). *For any $b > 0$ and any a,* (1) *is solvable, the optimal solution $q^* = q^*(a, b)$ is unique, and $q^*(a, b)$ is a locally Lipschitz mapping with respect to (a, b).*

2.2. A Bilevel Game

Now, the question is how profit-seeking generators and consumers behave under such a market structure. We use a model from Berry et al. (1999); c.f. Backerman et al. (2000) to address this question.

Assume that $A_i q_i + B_i q_i^2$ is the actual or true cost incurred by generator i to generate $q_i \geqslant 0$ units of electricity. In reality, this cost model does not capture the "lumpy" nature of generation costs due, for example, to start-up and ramping costs of individual units. Nevertheless, the quadratic representation is widely used, e.g., see Saadat (1999) and Alvarado et al. (2001). We use it because its simplicity enables further analysis, as in §3, and it captures some important aspects of generation, such as increasing marginal cost (for $A_i, B_i > 0$, assumed below). For consumer j, actual utility from consumption of quantity $-q_j \geqslant 0$ is modeled by $-A_j q_j - B_j q_j^2 = A_j(-q_j) - B_j(-q_j)^2$.

Throughout, we use capital letters for the true parameter values and the corresponding lower-case letters for the bid parameters. Given participant i's bid (a_i, b_i) and quantity q_i, the market price at node i is the marginal price, i.e., the derivative $a_i + 2b_i q_i$ of its bid function at q_i. Thus, a generator at node i receives revenue $(a_i + 2b_i q_i)q_i$, and its profit maximization problem is

$$\begin{aligned} \underset{a_i, b_i, q}{\text{maximize}} \quad & (a_i + 2b_i q_i)q_i - (A_i q_i + B_i q_i^2) \\ \text{subject to} \quad & \underline{A}_i \leqslant a_i \leqslant \bar{A}_i, \\ & \underline{B}_i \leqslant b_i \leqslant \bar{B}_i, \\ & q = (q_1, \ldots, q_i, \ldots, q_N) \\ & \qquad \text{solves (1) given } (a_{-i}, b_{-i}), \end{aligned} \tag{2}$$

where $a_{-i} = (a_1, \ldots, a_{i-1}, a_{i+1}, \ldots, a_N)$, $b_{-i} = (b_1, \ldots, b_{i-1}, b_{i+1}, \ldots, b_N)$; and $\underline{A}_i$ ($\underline{B}_i$) and $\bar{A}_i$ ($\bar{B}_i$) are the lower bound and upper bound on a_i (b_i), respectively. For the rest of this paper, we assume that $0 \leqslant \underline{A}_i \leqslant A_i \leqslant \bar{A}_i$ and $0 < \underline{B}_i \leqslant B_i \leqslant \bar{B}_i$.

Formally, (2) is called a bilevel program, where the lower-level program is the parametric problem (1); thus, $\{(2)\}_{i=1}^N$ is a bilevel game. In addition, because the lower-level problem is convex, we can represent it equivalently via its stationary conditions, which form an "equilibrium" system, as explained in §4.2. This converts the game to an EPEC; see Outrata (2003) for a general formulation of EPECs.

3. Nash Equilibria in the Electricity Market Model

As suggested in Berry et al. (1999) and shown explicitly in Example 4 to follow, the existence of pure-strategy Nash equilibria is not guaranteed for the electricity market game $\{(2)\}_{i=1}^N$. However, given Lemma 1, the existence of mixed-strategy Nash equilibria for $\{(2)\}_{i=1}^N$ is a straightforward consequence of a classical result (Balder 1995, Proposition 2.1) for games with finitely many players, each of whom has a continuous payoff function and compact set of pure strategies.

Our main interest is the existence of (pure-strategy) Nash equilibria of the game $\{(2)\}_{i=1}^N$: See also Berry et al. (1999), Younes and Ilic (1999), and Hobbs et al. (2000) for the model setting, choices of bidding parameters, or computational methods in special cases. In particular, we shall focus on bid-a-only and bid-b-only games for $\{(2)\}_{i=1}^N$—that is, leaving $b_i = B_i$ in the former and $a_i = A_i$ in the latter cases for $i = 1, \ldots, N$.

There are several reasons for this. First, the bid-a-only scenario provides a clear picture of how generators or

consumers shift their supply functions, $(p - a_i)/(2B_i)$, up or down in the oligopolistic environment. This is a convenient way to assess the extent of the existence of market power because a_i specifies the price at which generator i is willing to start its supply. Second, an equilibrium of the bid-b-only game is also an equilibrium of the bidding linear supply function in the sense of Klemperer and Meyer (1989) when there are no transmission constraints; see Hu et al. (2004) for a proof. In the latter case, generators are allowed to choose two parameters: both the intercept and the slope. Therefore, the bid-b-only game is naturally related to Klemperer and Meyer (1989) while providing an extension to a situation of limited transmission capacities. Third, we have found that a richer mathematical analysis is possible for the bid-a-only and bid-b-only scenarios (e.g., see §4.4) than for the bid-a-b scenario, which is related to our next point. Finally, the bid-a-b game has multiple equilibria in general, which makes economic comparisons fraught because it is not apparent which equilibrium to select, whereas both bid-a-only and bid-b-only games tend to have (locally) unique equilibria when these exist; see Hu (2003, Chapter 5). Nevertheless, §5 looks at all three game scenarios because bid-a-b games have a different numerical flavor from the others.

We present the game structure in §3.1 and give sufficient conditions for existence of Nash equilibria in §3.2. In §3.3, we give examples of existence, or lack thereof, of Nash equilibria. Our results here on the first aspect of our study are linked with the second aspect on alternative equilibrium concepts in §4.4 and in computational examples, namely, §5.2.3.

3.1. Notation for the Bid-*a*-Only Game Scenario

For $i = 1, \ldots, N$, b_i is fixed at its true value B_i. That is, we assume that B_i is known to the ISO and all players. In the context of player i, the multistrategy vector a may be written (a_i, a_{-i}). The optimal dispatch referred to in Lemma 1 will be denoted by $q^*(a)$ or $q^*(a_i, a_{-i})$; hence, player i's profit function is given by setting $b_i = B_i$ and $q_i = q_i^*(a_i, a_{-i})$ in the objective of (2):

$$f_i(a_i, a_{-i}) = (a_i - A_i)q_i^*(a_i, a_{-i}) + B_i q_i^*(a_i, a_{-i})^2. \tag{3}$$

This mapping is piecewise smooth on $\mathscr{A}$ by Lemma 1, where $\mathscr{A}_i = [\underline{A}_i, \bar{A}_i]$, $\mathscr{A} = \prod_{i=1}^N \mathscr{A}_i$. The ith player's profit or utility maximization problem is

$$\underset{a_i}{\text{maximize}}\, f_i(a_i, a_{-i}) \quad \text{subject to} \quad a_i \in \mathscr{A}_i. \tag{4}$$

This is called an *implicit program* (e.g., Luo et al. 1996) because the network and the associated pricing and dispatch problem are implicitly represented. We mention that (2) and (4) are equivalent in that a local or global minimizer of one is associated with a local or global minimizer, respectively, of the other. Similarly, the bilevel game $\{(2)\}_{i=1}^N$ and its implicit game formulation $\{(4)\}_{i=1}^N$ are equivalent in that a Nash equilibrium of one is associated with a Nash equilibrium of the other.

In this paper, we focus on games of complete information—that is, each player knows not only its own payoff functions, but other players' payoff functions, including their strategy spaces as well. It is worth noting that the above game $\{(4)\}_{i=1}^N$ has a classical format in that each player's strategy set is convex and compact, although the payoff functions are not smooth, but piecewise smooth. As we will see, the implicit payoff function $f_i(a_i, a_{-i})$ need not be concave in a_i, casting doubt on the existence of (pure-strategy) Nash equilibria. We also note, for future reference, that whatever bids a_{-i} are made by other players, player i's strategy can always be chosen to make his or her profit nonnegative: If $A_i \in \mathscr{A}_i$, then (3) gives $f_i(A_i, a_{-i}) \geqslant 0$.

3.2. Existence of Nash Equilibria

An often-cited sufficient condition for the existence of (pure-strategy) Nash equilibria is quasiconcavity of players' profit functions on their whole strategy spaces. However, Theorem 1 below assumes, for each player, nonemptiness and convexity of the subset of its strategies with nonnegative profit, and quasiconcavity of that player's profit function over this subset. We omit the proof, which is almost identical to that of the standard existence theorem; see, for example, Myerson (1991, pp. 138–140).

THEOREM 1. *Given a game with (a finite number) N of players for which the ith player has the payoff function $f_i(x_i, x_{-i})$ and the strategy set X_i, assume that the following conditions hold*:

(i) *the strategy space $X_i \subseteq \mathbb{R}^{n_i}$ is a nonempty compact set*;

(ii) *for each x_{-i} in $X_{-i} = \prod_{j:j\neq i} X_j$, the set $X_i(x_{-i}) := \{x_i \in X_i\colon f_i(x_i, x_{-i}) \geqslant 0\}$ is nonempty and convex*;

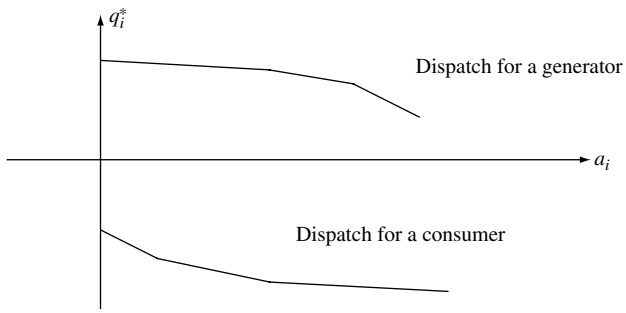
(iii) *for each $x_{-i} \in X_{-i}$, the function $f_i(\cdot, x_{-i})$ is quasiconcave on $X_i(x_{-i})$; and*

(iv) *$f_i(\cdot)$ is continuous in $\prod_{i=1}^N X_i$.*

Then, the game has at least one Nash equilibrium, such that every participant's profit is nonnegative.

To analyze the game $\{(4)\}_{i=1}^N$, we need some notation. Consider the mapping $a_i \mapsto q_i^*(a_i, a_{-i})$, which we denote $q_i^*(\cdot, a_{-i})$. By Lemma 1, there is a number W (depending on the number of pieces of the Karush-Kuhn-Tucker (KKT) system of (1)) such that, given a_{-i}, the derivative $q_i^{*\prime}(a_i, a_{-i})$ of $q_i^*(\cdot, a_{-i})$ exists except at the breakpoints $0 < w_1 = w_1(a_{-i}) \leqslant w_2 = w_2(a_{-i}) \leqslant \cdots \leqslant w_W = w_W(a_{-i})$. To explain further, associated with each w_k is at least one player j or line l such that in the former case the player is out of the market at $a_i = w_k$ ($q_j^*(a_i, a_{-i}) = 0$), but in the market for a_i either just below or just above w_k. Meanwhile, in the latter case, flow on link l is at full capacity when $a_i = w_k$, but drops below full capacity for a_i just below or just above w_k. We sketch the piecewise-linear dispatch function in Figure 1.

Figure 1. Sketch of the piecewise-linear dispatch for generators and consumers.



Given a_{-i}, let $\partial_i q_i^*(a_i, a_{-i})$ denote the generalized gradient (Clarke 1993) of $q_i^*(\cdot, a_{-i})$. (As an aside, we give the definition of a generalized gradient $\partial f(x^0)$ of an arbitrary locally Lipschitz function $f\colon \mathbb{R}^n \to \mathbb{R}$ at $x^0 \in \mathbb{R}^n$, using the classical result that the function is differentiable almost everywhere; see Clarke 1993 for details. The generalized gradient can be defined as the convex hull of limit points of sequences $\{\nabla f(x^k)\}$ over all sequences $\{x^k\}$ converging to x^0 such that $\nabla f(x^k)$ exists for each k. The local Lipschitz property ensures that the generalized gradient is nonempty and compact.) Our first result is technical. Its proof can be given from first principles; see the online appendix, which is available as part of the online version that can be found at http://or.journal.informs.org/.

Proposition 1. *For any participant $i = 1, \ldots, N$, its dispatch function $q_i^*(\cdot, a_{-i})$ is a nonincreasing funcstion of its own bid a_i given others' bid vector a_{-i}. Moreover, the directional derivative of $q_i^*(\cdot, a_{-i})$ along any direction $u \in \mathbb{R}$ satisfies*

$$|q_i^{*\prime}(a_i, a_{-i}; u)| \leqslant |u|/(2B_i).$$

As a result, $\partial_i q_i^*(a_i, a_{-i}) \subseteq [-1/(2B_i), 0]$.

As an application of Proposition 1, Lemma 1, and Theorem 1, we have the following theorem, whose proof is given in the online appendix. Condition (5) says that generators have decreasing marginal generation as their bid parameters increase, that is, each generator i's dispatch function is concave in a_i; see Figure 1. In other words, the extent of dependence of the ISO's dispatch on each generator is decreasing as the generator increases its bid. A similar interpretation holds for (6).

Theorem 2. *Suppose that player i's dispatch $q_i^*(\cdot, a_{-i})$ satisfies the following conditions for each a_{-i} and breakpoint w_j with $f_i(w_j, a_{-i}) > 0$:*

$$\textit{if } i \textit{ is a generator}\colon \quad \lim_{v \uparrow w_j} q_i^{*\prime}(v, a_{-i}) \geqslant \lim_{v \downarrow w_j} q_i^{*\prime}(v, a_{-i}), \tag{5}$$

$$\textit{if } i \textit{ is a consumer}\colon \quad \lim_{v \uparrow w_j} q_i^{*\prime}(v, a_{-i}) \leqslant \lim_{v \downarrow w_j} q_i^{*\prime}(v, a_{-i}), \tag{6}$$

where $v \uparrow w_j$ means $v \to w_j$, $v < w_j$; and $v \downarrow w_j$ is similarly defined. Then, the game $\{(4)\}_{i=1}^N$ has at least one Nash equilibrium such that each generator's profit and consumer's utility is nonnegative.

The next three results are applications of Theorem 2. In each case, conditions (5) and (6) can be established, yielding the existence of a Nash equilibrium. Such a Nash equilibrium may not be unique, but we can nevertheless show that, at a given Nash equilibrium, each generator with positive profit has a unique optimal response to the other player's equilibrium strategies.

First, we apply Theorem 2 to uncongested networks where all players participate in the market.

Proposition 2. *If for any $a \in \mathscr{A}$, the ISO's dispatch yields uncongested flows and is nonzero for each player, then there exists a Nash equilibrium for $\{(4)\}_{i=1}^N$. Moreover, given other players' strategies at such a Nash equilibrium, the optimal strategy of each player is unique.*

Proof. Because for any bid a, every generator or consumer has nonzero dispatch, the KKT conditions of the ISO's problem (1) contains the following linear system:

$$a_1 + 2B_1 q_1 + \lambda = 0, \ldots, a_N + 2B_N q_N + \lambda = 0,$$
$$q_1 + \cdots + q_N = 0,$$

where λ is the Lagrange multiplier corresponding to the constraint $q_1 + \cdots + q_N = 0$. As a result, $q_i^*(a_i, a_{-i})$ is a linear function for $(a_i, a_{-i}) \in \mathscr{A}$, so (5) and (6) hold trivially. Thus, by Theorem 2, there is a Nash equilibrium. Uniqueness of each player's best response to the others' strategies follows if we have strict concavity of profit functions. In fact, q_i^* is strictly decreasing in a_i (see (8) for its formula); therefore, by Proposition 1, we have $-1/(2B_i) \leqslant q_i^{*\prime} < 0$. Strict concavity results from (3). □

Second, we consider the case where consumers are symmetric (identical) and are price-takers (i.e., they do not bid; rather, their utility functions are known by all players and the ISO), but generators may have different generation costs. This weakens a typical assumption that generation costs are symmetric, e.g., Green and Newbery (1992) and Klemperer and Meyer (1989). Here, some generators may be out of the market, unlike in Proposition 2. Using symmetric consumer utilities ensures that all consumers have nonzero consumption, or all have zero consumption, which makes the mathematical analysis tractable.

Proposition 3. *Consider a network with capacities of all lines large enough so that congestion is not possible for any bids. Assume that consumers have the same utility function (i.e., they have the same A_i, B_i), which is known to the ISO and all players. Then, there exists at least one Nash equilibrium for the game $\{(4)\}_{i=1}^N$. Moreover, with respect to any such Nash equilibrium, the optimal strategy of each generator with a positive profit is unique.*

Proof. It suffices to establish (5) for a given generator i by fixing a_{-i} and a breakpoint $w = w_j$ of $q_i^*(\cdot, a_{-i})$ such that $q_i^*(w, a_{-i}) > 0$, and considering the derivative $q_i^{*\prime}(v, a_{-i})$ for v near w, but $v \neq w$.

In general, the vector (q, λ, ν) of optimal dispatches and the corresponding multipliers satisfies the following KKT conditions of (1) without line capacity constraints:

$$a_k + 2B_k q_k + \lambda + \nu_k = 0 \quad \text{for all } k,$$

$$q_k \nu_k = 0, \quad q_k \geqslant 0,\ \nu_k \leqslant 0, \quad k\text{: generator},$$

$$q_k \nu_k = 0, \quad q_k \leqslant 0,\ \nu_k \geqslant 0, \quad k\text{: consumer},$$

$$q_1 + \cdots + q_N = 0.$$

The KKT system can be used to derive the market price, which is uniform across the market, as

$$-\lambda(a_i, a_{-i}) = \left(\sum_{k\in K} \frac{a_k}{2B_k}\right) \Big/ \left(\sum_{k\in K} \frac{1}{2B_k}\right),$$

where $K = K(a_i, a_{-i}) = \{k\colon q_k^*(a_i, a_{-i}) \neq 0\}$ because $\nu_k = 0$ for $k \in K$. The KKT system then gives

$$-\lambda(a_i, a_{-i}) = a_k + 2B_k q_k^*(a_i, a_{-i}) \quad \text{for any } k \in K. \tag{7}$$

It follows that $q_i^*(\cdot, a_{-i})$ is differentiable at a_i if $i \in K$ and $K(v, a_{-i}) = K$ for v near a_i, with

$$q_i^{*\prime}(a_i, a_{-i}) = -(1 + L_i)/(2B_i),$$
$$\text{where } -1/L_i = \sum_{k\in K} B_i \Big/ B_k. \tag{8}$$

We need to establish some monotonicity properties. Suppose that the generator i has $i \in K = K(a_i, a_{-i})$. From (7) with $k = i$, the generalized gradient of the market price with respect to a_i is the set $1 + 2B_i \partial_i q_i^*(a_i, a_{-i})$, which contains only nonnegative values by Proposition 1. That is, price is nondecreasing in a_i. Next, consider a generator $k \notin K$, i.e., $q_k^*(a_i, a_{-i}) = 0$. We claim for $v < a_i$ that $q_k^*(v, a_{-i}) = 0$. For, if $q_k^*(v, a_{-i}) > 0$, then (7) at $a_i = v$ and the monotonicity of price imply $q_k^*(v, a_{-i})$ is nondecreasing as $v \uparrow a_i$; this can only happen, given the continuity of $q_k^*(\cdot, a_{-i})$ at a_i, if $q_k^*(v, a_{-i}) = 0$.

Now let w be a breakpoint as above. Then, $q_i^*(\cdot, a_{-i})$ is differentiable at v near w with $v \neq w$. Suppose that generator i unilaterally raises its linear bid by a small amount from w to $v > w$. By continuity of the vector $q^*(\cdot, a_{-i})$, each quantity $q_k^*(v, a_{-i})$ stays positive for $k \in K(w, a_{-i})$; hence,

$$K(v, a_{-i}) \supseteq K(w, a_{-i}) \quad \text{for } v > w,\ v \text{ near } w.$$

Suppose instead that player i unilaterally lowers its linear bid by a small amount to $v < w$. As before, we see that $K(v, a_{-i})$ contains $K(w, a_{-i})$, but we also claim the converse, namely,

$$K(v, a_{-i}) = K(w, a_{-i}) \quad \text{for } v < w,\ v \text{ near } w.$$

To see this, first note that because consumers are symmetric and total production is positive for v near w, each consumer k belongs to $K(v, a_{-i})$. Second, for a generator k with $q_k^*(w, a_{-i}) = 0$, we have already seen that $q_k^*(v, a_{-i}) = 0$ when $v < w$. The claimed equality follows. In light of these relationships and (8), with $a_i = v$ near but not equal to w and $K = K(v, a_{-i})$, inequality (5) can be easily verified.

The uniqueness claim follows from the fact that all these participants' profit functions are strictly concave when their strategies are restricted to those with strictly positive dispatch. □

Finally, we apply Theorem 2 to a two-node network with multiple generators and consumers at each node and a transmission capacity limit C on the only line between the two nodes. The line may be congested for some dispatches. This result has relevance, for example, in the New Zealand electricity market, which is a nodal pricing market where the South Island has 41% of the market generation capacity and 36% of market demand, whereas the North Island has the remainder of generation capacity and demand, and the two islands are connected by a single high-voltage direct current link. See David Butchers & Associates (2001) for more details for the New Zealand market.

Proposition 4. *Consider a network with two nodes and one line linking the two nodes with a transmission capacity limit $C > 0$ on the line. At each node, there may be multiple generators and/or consumers. If all consumers have the same utility function, which is known to the ISO and all generators, and they do not bid strategically, then there exists at least one Nash equilibrium for the game $\{(4)\}_{i=1}^N$. Moreover, with respect to any such Nash equilibrium, the optimal strategy of each generator with a positive profit is unique.*

Proof. The proof is almost the same as that of Proposition 3. For example, let i be a generator, a_{-i} be fixed, and $\mathcal{S} = \mathcal{S}(a_i, a_{-i})$ denote the set of generators k such that $q_k^*(a_i, a_{-i}) > 0$. The KKT conditions for this problem yield

$$-\lambda(a_i, a_{-i}) = \left(\sum_{k\in\mathcal{S}} \frac{a_k}{2B_k} + C\right) \Big/ \left(\sum_{k\in\mathcal{S}} \frac{1}{2B_k}\right),$$

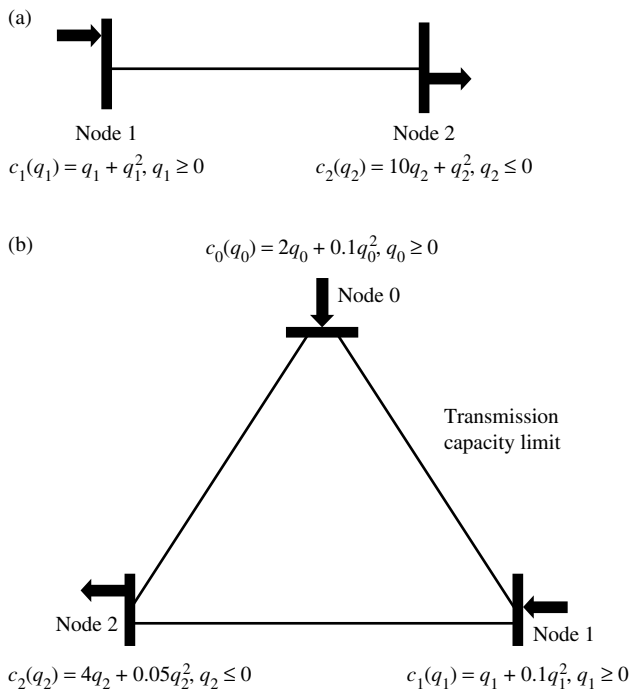
and $-\lambda(a_i, a_{-i}) = a_i + 2B_i q_i^*(a_i, a_{-i})$ if $i \in \mathcal{S}$. The proof can be completed as an exercise. □

3.3. Effects of Transmission Constraints on the Set of Nash Equilibria

In this section, we present two simple networks (see Figure 2) to see the effect of congestion on the existence of Nash equilibria of the bid-a-only game $\{(4)\}_{i=1}^N$. In the two-node case, the transmission limit leads to a continuum of Nash equilibria, whereas the transmission limit in a three-node network may result in the nonexistence of any Nash equilibrium.

The calculation of Nash equilibria for these two networks follows the classical idea of finding the intersection of every individual's best-response curves. The best-response function of participant i is a possibly set-valued mapping that maps a_{-i} to the set $\arg\max_{a_i}\{f_i(a_i, a_{-i})\colon a_i \in \mathcal{A}_i\}$. In the following examples, $\underline{A}_i = 0$, $\bar{A}_i = 100$.

Figure 2. Two networks, location of generators and consumers, and their cost and utility functions.



Note. Arrows show extraction or injection of electricity.

Example 1 (Two-Node Network, Uncongested). Consider a two-node network with one generator and one consumer and their true cost and utility functions shown in Figure 2(a).

When there is no capacity limit on the line, Proposition 2 tells that there exists a Nash equilibrium, and the equilibrium point is $(a_1^*, a_2^*) = (3.25, 7.75)$.

Example 2 (Two-Node Network, Congested). Now we consider the previous example with a capacity limit $C_{12} = 1$ on the line. Given a bid (a_1, a_2), the dispatches (a solution to problem (1)) are

$$q_1^* = -q_2^* = \frac{a_2 - a_1}{4} \quad \text{when } 0 \leqslant a_2 - a_1 \leqslant 4, \quad \text{or}$$
$$q_1^* = -q_2^* = 1 \quad \text{when } a_2 - a_1 \geqslant 4,$$

or $q_1^* = -q_2^* = 0$ otherwise. For further information, given any a_2, the breakpoints in $q^*(\cdot, a_2)$ occur at $a_1 = a_2 - 4$ ($q_1^* = 0$) and $a_1 = a_2$ ($q_1^* = 1$). For a_1 near the former, players 1 and 2 may both switch to being in or out of the market, whereas near the latter, flow on the line may switch between being congested and uncongested.

In fact, it can be shown that the best-response mappings for the generator and the consumer are

$$a_1^*(a_2) = \begin{cases} [a_2, 1] & \text{if } 0 \leqslant a_2 \leqslant 1, \\ \frac{1}{3}a_2 + \frac{2}{3} & \text{if } 1 \leqslant a_2 \leqslant 7, \\ a_2 - 4, & \text{if } a_2 \geqslant 7, \end{cases} \quad \text{and}$$

$$a_2^*(a_1) = \begin{cases} a_1 + 4, & \text{if } 0 \leqslant a_1 \leqslant 4, \\ \frac{1}{3}a_1 + \frac{8}{3}, & \text{if } 4 \leqslant a_1 \leqslant 10, \\ [0, a_1], & \text{if } a_1 \geqslant 10, \end{cases}$$

respectively. The intersection of the two best-response mappings gives the set of Nash equilibria for the game, which is $\{(a_1, a_2) \in \mathbb{R}^2\colon a_2 - a_1 = 4 \text{ and } 7 \leqslant a_2 \leqslant 8\}$.

In the previous example, and more generally, congested lines may lead to multiplicity of local Nash equilibria. The multiplicity may cause ambiguity of economic conclusions based on the equilibria. See debates on this between Stoft (1999) and Oren (1997) for a Cournot-Nash model representing an electricity market with two-node and three-node networks. In general, we have the following corollary, which is different from Proposition 4 in that the consumer in Corollary 1 participates in strategic bidding, and multiple generators are allowed in Proposition 4.

Corollary 1. *Consider any two-node, one-line network with only one generator and one consumer located at different nodes. There always exists at least one Nash equilibrium for* $\{(4)\}_{i=1}^2$ *regardless of capacity limit on the line.*

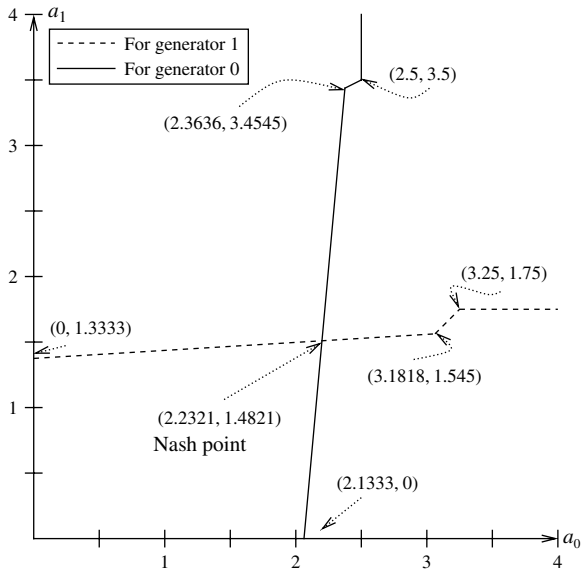
Proof. If there is no capacity limit on the line, the corollary is true by Proposition 2. Now assume that there is a capacity limit C on the line. Consider condition (6) for the consumer (a similar proof holds for the generator). Fix the generator's bid $a_1 \geqslant 0$, if the capacity limit on the line is just reached by a dispatch given bid (a_1, w); i.e., $q_1^* = -q_2^* = C$ when $a_2 \geqslant w$ and $q_1^* = -q_2^* < C$ when $a_2 < w$, where w is a bid by the consumer. Therefore, $q_2^{*\prime} = 0$ for $a_2 > w$ because $q_2^*(a_1, a_2) = -C$ for $a_2 \geqslant w$. Similarly, $q_2^{*\prime} \leqslant 0$ for $a_2 < w$ (see also Proposition 1). Therefore, (6) holds for the consumer. Hence, the corollary follows from Theorem 2. □

Example 3 (Three-Node Network, Uncongested). Consider a three-node network given in Figure 2(b). There are two competing generators, and one consumer who does not compete, but rather reveals her true utility function to the ISO and the two generators. The true cost and utility functions are shown in Figure 2(b). Node 2 is chosen as the reference node, and therefore its distribution factors to all three lines are zero. The distribution factors of node 0 to lines 0–1, 0–2, 1–2 are −0.25, −0.75, −0.25, respectively, and those of node 1 to the above lines are 0.25, −0.25, −0.75. That is, line 0–1 transmits electricity twice as easily as line 0–2 and line 1–2, or it has half the susceptance of the other lines.

Without any transmission limits on the three lines, we know by Proposition 3 that $\{(4)\}_{i=1}^2$ has a Nash equilibrium, which is $(a_1^*, a_2^*) = (2.2321, 1.4821)$ shown in Figure 3.

Example 4 (Three-Node Network, Congested). Next, assume that there is a capacity limit $C_{01} = 0.9$ on the line

Figure 3. Best-response curves without transmission limit.



between node 0 and node 1. There are only two possible nontrivial cases, i.e., with both generators having positive generation, coming out of the KKT system of (1):

$$q_0^* = 10 - 3.75a_0 + 1.25a_1, \quad q_1^* = 10 + 1.25a_0 - 3.75a_1,$$
$$q_2^* = -20 + 2.5a_0 + 2.5a_1,$$

when $a_0 - a_1 \leqslant 0.72$ (uncongested situation) or

$$q_0^* = 8.2 - 1.25a_0 - 1.25a_1, \quad q_1^* = 11.8 - 1.25a_0 - 1.25a_1,$$
$$q_2^* = -20 + 2.5a_0 + 2.5a_1$$

when $a_0 - a_1 \geqslant 0.72$ (congested situation).

It is easy to see that condition (5) in Theorem 2 is violated for generator 0:

$$q_0^{*\prime}(v, a_1)|_{v<a_1+0.72} = -3.75 \ngeqslant -1.25 = q_0^{*\prime}(v, a_1)|_{v>a_1+0.72}.$$

Therefore, Theorem 2 cannot apply to the game in this example.

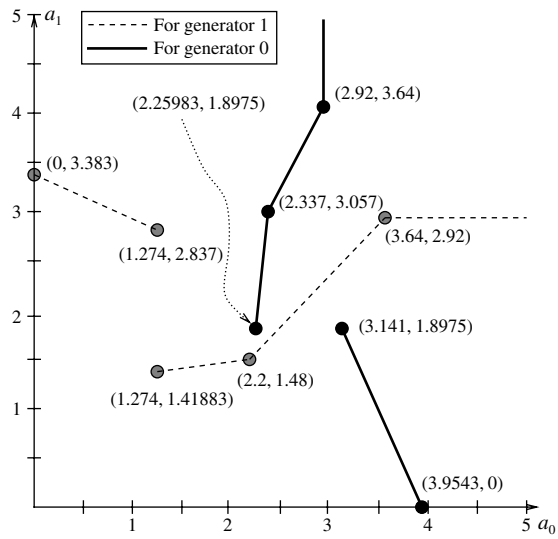
In fact, the best-response functions for the two generators are

$$a_0^*(a_1) = \begin{cases} 3.9543 - 0.4286a_1 & \text{if } 0 \leqslant a_1 \leqslant 1.8975, \\ 0.0667a_1 + 2.1333 & \text{if } 1.8975 \leqslant a_1 \leqslant 3.057, \\ a_1 - 0.72 & \text{if } 3.057 \leqslant a_1 \leqslant 3.64, \\ 2.92 & \text{if } a_1 \geqslant 3.64, \end{cases}$$

and

$$a_1^*(a_0) = \begin{cases} 3.383 - 0.4286a_0 & \text{if } 0 \leqslant a_0 \leqslant 1.274, \\ 0.0667a_0 + 1.3333 & \text{if } 1.274 \leqslant a_0 \leqslant 2.2, \\ a_0 - 0.72 & \text{if } 2.2 \leqslant a_0 \leqslant 3.64, \\ 2.92 & \text{if } a_0 \geqslant 3.64. \end{cases}$$

Figure 4. Best-response curves with one transmission limit.



It can be clearly seen that there is no Nash equilibrium for the game in this example from Figure 4 because there is no intersection point for their best-response curves.

4. EPEC Formulations of Bilevel Games

This section is devoted to the second aspect of our study, in particular, reformulating bilevel games as EPECs and thereby considering equilibrium concepts that are alternatives to the Nash equilibrium. We start with a bilevel game model that generalizes the above electricity market models and, by writing each player's problem as an MPEC, reformulate the game as an EPEC; see §4.1. In §4.2, we propose *local Nash* and *Nash stationary equilibria* as more tractable concepts than the standard Nash equilibria. Nash stationary equilibria can be posed as solutions of a mixed complementarity problem, which is the subject of §4.3. Finally, §4.4 returns to the electricity market game $\{(4)\}_{i=1}^N$ to give conditions under which Nash stationary points are, in fact, local Nash equilibria.

4.1. A General Bilevel Game and Its EPEC Formulation

Given x_{-i}, player i's problem in the general bilevel game is to

$$\begin{aligned} &\underset{x_i}{\text{maximize}} \quad \phi_i(x, y) \\ &\text{subject to} \quad g_i(x_i, y) \geqslant 0, \quad h_i(x_i, y) = 0, \\ &\qquad y \text{ solves} \begin{cases} \underset{y}{\text{minimize}} \quad F(x, y) \\ \text{subject to} \quad H(x, y) = 0, \\ \qquad\qquad G(x, y) \geqslant 0, \end{cases} \end{aligned} \tag{9}$$

where $x = (x_1, \ldots, x_N) \in \mathbb{R}^{n_1+\cdots+n_N}$, $y = (y_1, \ldots, y_N) \in \mathbb{R}^{m_1+\cdots+m_N}$; the scalar-valued functions ϕ_i, F are smooth; and g_i, h_i, G, and H are smooth vector functions of dimensions w_i, v_i, s, and t, respectively, for $i = 1, \ldots, N$. Obviously, (9) generalizes the bilevel game (2) by taking $x = (a, b)$, $x_i = (a_i, b_i)$, $y = q$, and appropriate forms of the functions ϕ_i, g_i, etc. It also generalizes the bid-a-only version of (2) represented by the implicit program (4).

Given x, it is to be expected that a solution y of the lower-level problem in (9) is a stationary point of the lower-level problem, i.e., it solves the associated KKT system, which involves multipliers $\lambda \in \mathbb{R}^s$ and $\mu \in \mathbb{R}^t$:

$$\begin{aligned} &\nabla_y F(x, y) - \nabla_y G(x, y)^T \mu + \nabla_y H(x, y)^T \lambda = 0, \\ &H(x, y) = 0, \qquad 0 \leqslant G(x, y) \perp \mu \geqslant 0, \end{aligned} \tag{10}$$

where $\perp$ denotes orthogonality: $G(x, y)^T \mu = 0$. The vectors λ, μ are variously called Lagrange multipliers, KKT multipliers, or shadow prices.

Within the bilevel program (9), replacing the lower-level optimization problem by (10) creates an MPEC; indeed, a mathematical program with complementarity constraints or MPCC:

$$\begin{aligned} \underset{x_i, y, \mu, \lambda}{\text{maximize}} \quad & \phi_i(x, y) \\ \text{subject to} \quad & g_i(x_i, y) \geqslant 0, \quad h_i(x_i, y) = 0, \\ & (y, \lambda, \mu) \text{ solves (10).} \end{aligned} \tag{11}$$

The corresponding EPEC is the game $\{(11)\}_{i=1}^N$; it might well be called an equilibrium program with complementarity constraints (EPCC). An attraction of reformulating a bilevel program as an MPEC is that the latter has an explicit structure; indeed, we will see that it can easily be phrased as a nonlinear program, and hence nonlinear programming technology can be applied.

It can be observed that bilevel programs and MPECs typically have nonconvex feasible sets; ergo, the corresponding games have nonconvex strategy sets. We can view this difficulty through a different lens by supposing that the lower-level solution y is uniquely determined as $y(x)$ by the upper-level variables x. Here, player i's implicit payoff function $\phi_i(x, y(x))$ is typically nonconcave in x_i, which casts doubt on the existence of Nash equilibria. This is precisely the situation discussed in §3 for the implicit bid-a-only game $\{(4)\}_{i=1}^N$. Thus, our computational strategy, rather than attempting to demonstrate or disprove the existence of a Nash equilibrium, will be to apply robust mathematical programming software that aims to solve certain necessary conditions for Nash equilibria. That is, we will search for joint strategies x and corresponding lower-level solutions $y(x)$ such that for each i, $(x_i, y(x))$ is stationary for the MPEC (11) and, if so, then check whether this point is a local solution of the MPEC.

Recall Example 2, the bid-a-only game on a 2-node network with a capacitated link. We will return to this several times, in Examples 2A–2C, developed to illustrate some concepts and results.

Example 2A. We use the format of (11) to model the bid-a-only game of Example 2, in which $A_1 = 1$, $B_1 = 1$ and $A_2 = 10$, $B_2 = 1$; see Figure 2(a). The ISO's problem, with $b_i = B_i$ for $i = 1, 2$, is

$$\begin{aligned} \underset{q_1, q_2}{\text{minimize}} \quad & a_1 q_1 + q_1^2 + a_2 q_2 + q_2^2 \\ \text{subject to} \quad & q_1 + q_2 = 0 \quad : \quad \mu, \\ & q_1 \leqslant 1 \quad : \quad \nu_1, \\ & q_1 \geqslant 0 \geqslant q_2 \quad : \quad \nu_2, \end{aligned}$$

where KKT multipliers appear at the right. In the bilevel game, player 1, the generator, is given a_2 and wishes to maximize $(a_1 - 1)q_1 + q_1^2$ by choosing a_1, without lower and upper bounds in this example, where q_1 as well as q_2 must solve the ISO's problem above. Similarly, the consumer, given a_1, wishes to maximize $(a_2 - 10)q_2 + q_2^2$ by choosing a_2 subject to q_1 and q_2 solving the ISO's problem.

To form the EPEC, first define the associated Lagrangian function as

$$\begin{aligned} L(q_1, q_2, \lambda, \mu, \nu) = a_1 q_1 + q_1^2 + a_2 q_2 + q_2^2 + \lambda(q_1 + q_2) \\ - \mu(1 - q_1) - q_1 \nu_1 + \nu_2 q_2. \end{aligned}$$

We write the KKT conditions in two parts. First,

$$\begin{aligned} &0 = \nabla_{q_1} L = a_1 + 2q_1 + \lambda + \mu - \nu_1, \\ &0 = \nabla_{q_2} L = a_2 + 2q_2 + \lambda + \nu_2, \\ &0 = q_1 + q_2, \\ &0 \leqslant 1 - q_1, \quad \mu \geqslant 0, \\ &0 \leqslant q_1, \quad \nu_1 \geqslant 0, \\ &0 \geqslant q_2, \quad \nu_2 \geqslant 0, \end{aligned} \tag{12}$$

and second, orthogonality conditions: $1 - q_1 \perp \mu$, $q_1 \perp \nu_1$, and $q_2 \perp \nu_2$. In the EPEC, player 1 is given a_2 and maximizes $(a_1 - 1)q_1 + q_1^2$ by choosing a_1, where q_1, q_2, λ, μ, ν_1, and ν_2 must satisfy the above KKT conditions, whereas player 2, given a_1, maximizes $(a_2 - 10)q_2 + q_2^2$ by choosing a_2 subject to the KKT conditions on the remaining variables.

The Nash points found in Example 2 are characterized by $a_1 = a_2 - 4$, $7 \leqslant a_2 \leqslant 8$. In each case, the corresponding quantities and multipliers are uniquely defined as $q_1 = 1 = -q_2$ and $\mu = \nu_1 = \nu_2 = 0$.

4.2. Local Nash Equilibria and Nash Stationary Equilibria

A local solution of (11), given $x_{-i} = x_{-i}^*$, is a point $(x_i^*, y^*, \lambda^*, \mu^*)$ for which there exists a neighborhood of this point, denoted by $\mathcal{N}_i$, such that $(x_i^*, y^*, \lambda^*, \mu^*)$ is optimal for (11) with the extra constraint $(x_i, y, \lambda, \mu) \in \mathcal{N}_i$.

Definition 1. A strategy vector $(x^*, y^*, \lambda^*, \mu^*)$ is called a *local Nash equilibrium* for the EPEC $\{(11)\}_{i=1}^N$ if, for each i, $(x_i^*, y^*, \lambda^*, \mu^*)$ is locally optimal for the MPEC (11) when $x_{-i} = x_{-i}^*$.

We may refer to a global Nash equilibrium (meaning a Nash equilibrium), to distinguish it from a local Nash equilibrium. While local Nash equilibria seem deficient relative to their global counterparts, we argue that they have some meaning. Local optimality may be sufficient for the satisfaction of players, given that global optima of nonconcave maximization problems are difficult to identify. Putting it differently, limits to rationality or knowledge of players may lead to meaningful local Nash equilibria. For example, generators may only optimize their bids locally (via small adjustments) due to operational constraints or general conservativeness. Another limit to rationality is implicit in the simple nature of the model $\{(11)\}_{i=1}^N$ explored in §3, e.g., a DC model of an electricity network is an approximation derived by linearizing a nonlinear AC power flow model in a steady state. Because very different bids may be better studied using different DC approximations, it is not clear how much more attractive a global Nash equilibrium would be relative to a local Nash equilibrium of this model.

A further challenge, computationally speaking, is how to find a local optimizer for a single player. We adopt the classical strategy from nonlinear optimization in which a system of stationary conditions that necessarily holds at a locally optimal point is the main criterion for recognizing local optimality. Hence, we further relax the equilibrium solution concept to allow for a joint strategy $(x^*, y^*, \lambda^*, \mu^*)$ such that each $(x_i^*, y^*, \lambda^*, \mu^*)$ is stationary, in a sense that we define next, for player i's MPEC.

Given x_{-i}, the MPEC (11) is equivalent to a nonlinear program (NLP):

$$\begin{aligned}
\underset{x_i, y, \lambda, \mu}{\text{maximize}} \quad & \phi_i(x, y) \\
\text{subject to} \quad & g_i(x_i, y) \geqslant 0\colon\ \xi_i^g \geqslant 0, \\
& h_i(x_i, y) = 0\colon\ \xi_i^h, \\
& \nabla_y F(x, y) - \nabla_y G(x, y)^T \mu \\
& \quad + \nabla_y H(x, y)^T \lambda = 0\colon\ \eta_i, \\
& H(x, y) = 0\colon\ \xi_i^H, \\
& G(x, y) \geqslant 0\colon\ \xi_i^G \geqslant 0, \\
& \mu \geqslant 0\colon\ w_i \geqslant 0, \\
& G(x, y) * \mu = 0\colon\ \zeta_i,
\end{aligned} \tag{13}$$

where $*$ denotes the Hadamard or componentwise product of vectors, i.e., $G(x, y) * \mu = (G_1(x, y)\mu_1, \ldots, G_s(x, y)\mu_s)$. The vectors at right—$\xi_i^g$, etc.—are multipliers that will be used later.

Thanks to recent developments on MPECs, we will see that the following stationary condition, which is developed further in §4.3, is sensible for the EPEC $\{(11)\}_{i=1}^N$. In the definition, we use the term "point" when we could equally well use "strategy" or "equilibrium."

Definition 2. A strategy $(x^*, y^*, \lambda^*, \mu^*)$ is called a *Nash stationary point* of the EPEC $\{(11)\}_{i=1}^N$ if, for each $i = 1, \ldots, N$, the point $(x_i^*, y^*, \lambda^*, \mu^*)$ is stationary for (13) with $x_{-i} = x_{-i}^*$.

To our knowledge, the idea of Nash stationary points has not been formalised previously. However, it is certain that Nash stationary points have been used in modeling and computational approaches to classical games; see, e.g., Facchinei and Pang (2003). What may be novel is to approach stationarity of bilevel programs, hence bilevel games, via MPECs.

The definition of Nash stationary points is justified by Proposition 5 (to follow) under the following constraint qualification. First, recall that, given a feasible point of a system of equality and inequality constraints, an active constraint is a scalar-valued constraint that is satisfied as an equality.

Definition 3. Let (x, y, λ, μ) be feasible for the EPEC $\{(11)\}_{i=1}^N$.

(i) Given a player index i, the *MPEC active constraints* at (x_i, y, λ, μ) are the active constraints of (11) ignoring the orthogonality condition denoted by $\perp$ (equivalently, the active constraints of (13) excluding the last block equation $G(x, y) * \mu = 0$).

(ii) *MPEC linear independence constraint qualification*, MPEC-LICQ, holds for (11) at (x_i, y, λ, μ) if the gradients, with respect to these variables, of the MPEC active constraints at the point (x, y, λ, μ), are linearly independent.

(iii) We say that the MPEC-LICQ holds for $\{(11)\}_{i=1}^N$ at (x, y, λ, μ) if, for each player i, the MPEC-LICQ holds for (11) at (x_i, y, λ, μ).

Linear independence of active gradients at a local minimum of a traditional nonlinear program means that the associated KKT multipliers or shadow prices are unique, e.g., in economic terms, if constraints represent resource limits, then the value of additional resource is (uniquely) priced.

Proposition 5. *Let $(x^*, y^*, \lambda^*, \mu^*)$ be feasible for the EPEC $\{(11)\}_{i=1}^N$, and consider the statements*:

(i) *(x^*, y^*) is a local Nash equilibrium of the bilevel game $\{(9)\}_{i=1}^N$.*

(ii) *$(x^*, y^*, \lambda^*, \mu^*)$ is a local Nash equilibrium of the EPEC $\{(11)\}_{i=1}^N$.*

(iii) *$(x^*, y^*, \lambda^*, \mu^*)$ is a Nash stationary point of the EPEC $\{(11)\}_{i=1}^N$.*

Statement (i) *implies statement* (ii). *Statement* (ii) *implies statement* (iii) *if the MPEC-LICQ holds for the EPEC at $(x^*, y^*, \lambda^*, \mu^*)$.*

The proof of this proposition is available in the online appendix.

Example 2B. We continue Example 2A by looking at how the MPEC-LICQ applies. Cast player 1's problem as in the style of (13): Given a_2, choose $(a_1, q_1, q_2, \lambda, \mu, \nu_1, \nu_2)$ to maximize $(a_1 - 1)q_1 + q_1^2$ subject to, first, (12) and, second, the bilinear equations $0 = (1 - q_1)\mu$, $0 = q_1\nu_1$, and $0 = q_2\nu_2$. Likewise, player 2, given a_1, chooses $(a_2, q_1, q_2, \mu, \lambda, \nu_1, \nu_2)$ to maximize $(a_2 - 10)q_2 + q_2^2$ subject to these constraints.

At a feasible point with $q_1 \neq 0$ (equivalently, $q_2 \neq 0$), the MPEC active constraints are given by the list

$$
\begin{aligned}
0 &= a_1 + 2q_1 + \lambda + \mu - \nu_1,\\
0 &= a_2 + 2q_2 + \lambda + \nu_2,\\
0 &= q_1 + q_2,\\
0 &= q_1 - 1,\\
0 &= \mu,\\
0 &= \nu_1,\\
0 &= \nu_2.
\end{aligned}
$$

Imagine perturbing the left-hand zeros to any numbers. A solution can be found easily by looking at these equations in reverse order, i.e., the system has full column rank. Equivalently, this system has full row rank, i.e., these equations have linearly independent gradients and the MPEC-LICQ holds.

A similar argument shows that the MPEC-LICQ holds at any solution of player 2's problem for which $q_1 \neq 0$. Hence, Proposition 5 says that any such point is also Nash stationary for this game.

Remarks. (i) NLP reformulations of MPECs, as above, are unusually difficult in that they do not satisfy constraint qualifications that are standard in NLP theory. This difficulty is alleviated by the MPEC-LICQ. Without the MPEC-LICQ, however, it is possible to have a local minimum of (13) that is not stationary—hence, a local or global Nash equilibrium of the associated EPEC $\{(11)\}_{i=1}^N$ that is not Nash stationary.

(ii) According to Scholtes and Stöhr (2001), the MPEC-LICQ holds generically for MPECs. For example, for an MPEC with a quadratic objective function and linear constraints, almost all (with respect to Lebesgue measure) data matrices and vectors used to specify the problem will give an MPEC for which the MPEC-LICQ holds at all feasible points. This result, combined with the above proposition, does not promise, but does give us reasonable hope of finding Nash points, if they exist, by identifying Nash stationary points.

(iii) Stationarity of $(x_i^*, y^*, \lambda^*, \mu^*)$ for (13), where $x_{-i} = x_{-i}^*$, is equivalent by Anitescu (2005) to a kind of MPEC stationarity called *strong stationarity* (Scheel and Scholtes 2000). Fletcher et al. (2006) give this result for another nonlinear programming formulation of MPECs. To this extent, we could refer to Nash stationary points as Nash strongly stationary points.

4.3. Complementarity Problem Formulation

We turn to a CP form of the Nash stationary point conditions of an EPEC. This is an aggregation of the KKT systems of (13) for each player. We call this the *complementarity problem* (*CP*) *formulation of the EPEC* $\{(11)\}_{i=1}^N$; it is referred to in Hu (2003) and Hu et al. (2004) as the *All-KKT system*.

The Lagrangian function for player i's MPEC (13) is, omitting arguments (x, y) at right,

$$
\begin{aligned}
&L_i(x, y, \lambda, \mu, \xi_i^g, \xi_i^h, \eta_i, \xi_i^G, \xi_i^H, \zeta_i)\\
&\quad = \phi_i + g_i^T\xi_i^g + h_i^T\xi_i^h + (\nabla_y F - \nabla_y G^T\mu + \nabla_y H^T\lambda)^T\eta_i\\
&\qquad + \mu^T w_i + H^T\xi_i^H + G^T\xi_i^G + (G * \mu)^T\zeta_i.
\end{aligned}
$$

The CP formulation of the EPEC $\{(11)\}_{i=1}^N$ is

$$
\left.\begin{aligned}
&0 = \nabla_{(x_i, y, \lambda, \mu)} L_i,\\
&0 \leqslant \xi_i^g \perp g_i(x_i, y) \geqslant 0,\\
&0 = h_i(x_i, y),\\
&0 \leqslant \xi_i^G \perp G(x, y) \geqslant 0,\\
&0 \leqslant \mu \perp w_i \geqslant 0,
\end{aligned}\right\} \quad \forall i, \tag{14.1}
$$

$$
\left.\begin{aligned}
&0 = \nabla_y F(x, y) - \nabla_y G(x, y)^T\mu + \nabla_y H(x, y)\lambda,\\
&0 = H(x, y),\\
&0 = G(x, y) * \mu,
\end{aligned}\right\} \tag{14.2}
$$

where ξ_i^g, ξ_i^h, ξ_i^H, ξ_i^G, η_i, w_i, and ζ_i are the KKT multiplier vectors that are listed in the right-most column in (13). All players share the second group of constraints, (14.2); there is no need for it to appear more than once in the CP formulation. (This is called a "mixed" CP because equality constraints appear.)

Note that by ordering the vectors $(x_i, \eta_i, \xi_i^H, \zeta_i)$, ξ_i^g, ξ_i^h, ξ_i^G, and w_i for all i, we match their respective dimensions to the number of equations or complementarity relations of the blocks of (14.1), and likewise, y, λ, μ match the blocks of (14.2). That is, Equations (14.1) and (14.2) define a complementarity problem to which a standard solver like PATH (Dirkse and Ferris 1995, Ferris and Munson 1999) can be applied. See Ehrenmann (2004a) for an alternative CP formulation.

Applying PATH to the CP formulation of an EPEC is suggested by the success that some standard NLP methods have had in solving MPECs written as NLPs (like (13)). The sequential quadratic programming (SQP) method has proven to be especially effective (Anitescu 2005, Fletcher and Leyffer 2004, Fletcher et al. 2006). This particularly motivates the application of PATH because its underlying methodology (Ralph 1994, Robinson 1994) is the analog, for CPs, of SQP.

4.4. Local Nash Equilibria of Bid-*a*-Only Games

Given a Nash stationary point, one would like to check a further optimality condition to see if it is actually a (local) Nash equilibrium. For example, a stationary point of a standard nonlinear program is an isolated local minimizer if a second-order sufficient condition holds at that point; see Bonnans and Shapiro (1998). Here we revisit electricity market games, namely, the bid-a-only implicit program $\{(4)\}_{i=1}^N$, to show that Nash stationary points are local Nash points under reasonable conditions.

For brevity, we will not address the bid-b-only scenario, but refer the reader to Hu (2003), which shows how to check second-order sufficient optimality conditions via second-order parabolic directional derivatives (Ben-Tal and Zowe 1982) of the objective function, and gives preliminary numerical results for which this checking usually verifies local optimality of each player's equilibrium bid.

To speak about Nash stationary equilibria for the bid-a-only game, we need to define the MPEC faced by player i, which is derived from (2) by setting $b_i = B_i$ and replacing the condition that q solves the ISO's pricing and dispatch problem (1) by the KKT conditions of the latter:

$$\begin{aligned}
\underset{a_i,\, q,\, \lambda,\, \underline{\mu},\, \bar{\mu},\, \nu}{\text{maximize}} \quad & (a_i - A_i)q_i + B_i q_i^2 \\
\text{subject to} \quad & \underline{A}_i \leqslant a_i \leqslant \bar{A}_i, \\
& 0 = q_1 + \cdots + q_N, \\
& \left.\begin{aligned}
& 0 = a_j + 2B_j q_j + \lambda + \sum_{l=\mathscr{L}} \Phi_{l,j}(\bar{\mu}_l - \underline{\mu}_l) - \nu_j, \\
& 0 \leqslant q_j \perp \nu_j \geqslant 0 \quad \text{if player } j \text{ is a generator}, \\
& 0 \geqslant q_j \perp \nu_j \leqslant 0 \quad \text{if player } j \text{ is a consumer},
\end{aligned}\right\} \\
& \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad j = 1, \ldots, N, \\
& \left.\begin{aligned}
& 0 \leqslant C_l + \sum_{k=1}^N \Phi_{l,k} q_k \perp \underline{\mu}_l \geqslant 0, \\
& 0 \leqslant C_l - \sum_{k=1}^N \Phi_{l,k} q_k \perp \bar{\mu}_l \geqslant 0,
\end{aligned}\right\} \quad l \in \mathscr{L}.
\end{aligned} \tag{15}$$

Denote by $\underline{\mu}$, $\bar{\mu}$, q, and ν the vectors whose respective components are $\underline{\mu}_l$, $\bar{\mu}_l$, q_j, and ν_j; these are the KKT multipliers of (1).

The next result is a kind of converse to Proposition 5. Its main conclusion is that a Nash stationary point of the bid-a-only electricity game is actually a local Nash equilibrium, provided that the KKT multipliers of the lower-level problem (1) depend uniquely on the bids. This strong result relies on the particular structure of (4) or (15). To obtain uniqueness of lower-level multipliers, we assume that the active gradients of (1) are linearly independent at the point in question, namely, the LICQ holds.

Theorem 3. *Let $(a^*, q^*, \lambda^*, \underline{\mu}^*, \bar{\mu}^*, \nu^*)$ be a Nash stationary point for the EPEC $\{(15)\}_{i=1}^N$. If the LICQ holds for the ISO's pricing and dispatch problem* (1) *at $q = q^*$, with $a = a^*$, then*

(i) *$(a^*, q^*, \lambda^*, \underline{\mu}^*, \bar{\mu}^*, \nu^*)$ is a local Nash equilibrium of the bid-a-only EPEC $\{(15)\}_{i=1}^N$;*

(ii) *(a^*, q^*) is a local Nash equilibrium of the associated bid-a-only bilevel game, i.e., $\{(2)\}_{i=1}^N$, where each player i sets $b_i = B_i$; and*

(iii) *a^* is a local Nash equilibrium for the bid-a-only implicit game $\{(4)\}_{i=1}^N$.*

Moreover, each generator's profit or consumer's utility $f_i(a_i^*, a_{-i}^*)$ is nonnegative.

Example 2C. Recall Example 2, and the associated bilevel game and EPEC given in Example 2A. The ISO's dispatch problem has constraints $q_1 + q_2 = 0$, $q_1 \leqslant 1$, and q_1, q_2 nonnegative. Thus, provided $q_1 \neq 0$ or, equivalently, $q_2 \neq 0$, the active constraints consist of either the first constraint or the first two constraints. In both cases, the active constraint gradients are linearly independent, i.e., LICQ holds there. Theorem 3 now says that any Nash stationary point $(a_1, a_2, q_1, q_2, \mu, \lambda, \nu_1, \nu_2)$ of the EPEC is also a local Nash equilibrium. Likewise, (a_1, a_2, q_2, q_2) and (a_1, a_2) are local Nash equilibria of the corresponding bilevel game and implicit game, respectively.

Conversely, suppose that we have a local Nash equilibrium of the implicit game. This yields a local Nash equilibrium for the bilevel game (as noted in the discussion following the formulation of (4)) and, from Proposition 5, a Nash equilibrium for the EPEC. If, in addition, the quantity q_1 is nonzero, then Example 2B tells us that the EPEC solution is Nash stationary.

In summary, for this simple electricity market over two nodes, Nash stationary points of the EPEC and local Nash equilibria of the EPEC or bilevel program or implicit program are all equivalent concepts, provided both players are in the market.

The proof of Theorem 3, given in the online appendix, uses optimality conditions for nonsmooth problems like (4) that we will sketch below. The burden of the proof is in two parts. The first is to show that Nash stationary points imply a kind of nonsmooth stationarity for each player's implicit program (4). This result holds in a certain generality, which is the content of Proposition 6 below. We then show that nonsmooth stationarity of (4) for each player is sufficient for local optimality, a property that relies on the particular structure of this game.

It will be convenient to use the general bilevel profit maximization (9) under some simplifying assumptions. The objective ϕ_i is required to be smooth, as previously, and we further assume that

(a) each of the upper-level constraint mappings g_i and h_i is an affine function of x_i only;

(b) F is quadratic in (x, y) and strictly convex in y for each fixed x; and

(c) G and H are affine.

A consequence of assumptions (b) and (c) is that the lower-level solution y is uniquely defined by x as a piecewise-linear function $y(x)$; cf. Lemma 1. (If, in addition, the LICQ holds analogous to Theorem 3, then the multipliers $\lambda(x)$, $\mu(x)$ exist uniquely as piecewise-smooth functions of x near x^*.) Similar to our formulation for the bid-a-only game in §3.1, uniqueness of $y(x)$ allows us to reformulate each player's problem (9) as an implicit program (Luo et al. 1996): Let $f_i(x_i, x_{-i}) = \phi_i(x, y(x))$ and consider

$$\begin{aligned} &\underset{x_i}{\text{maximize}} \quad f_i(x_i, x_{-i}) \\ &\text{subject to} \quad g_i(x_i) \geqslant 0, \quad h_i(x_i) = 0. \end{aligned} \tag{16}$$

The constraints are now linear or polyhedral. The objective function is the composition of a smooth mapping with a piecewise-linear mapping; hence, it is piecewise smooth and both locally Lipschitz and directionally differentiable. The last two properties together define what is called B-differentiability (Shapiro 1990). We denote its partial directional derivative with respect to x_i in a direction d_i (of the same dimension n_i) by $f_i'(x_i, x_{-i}; d_i)$.

Suppose that x_{-i} is given as x_{-i}^* and x_i^* is feasible for (16). A tangent direction to the feasible set at x_i^* is a vector d_i that can be written as the limit, as $k \to \infty$, of a convergent sequence $(x_i^k - x_i^*)/\tau_k$, where each x^k is feasible and each τ_k is a positive scalar. (Because the feasible set is polyhedral, the tangent directions are precisely positive scalings of vectors $x_i - x_i^*$, where x_i is feasible and near x_i^*.) If x_i^* is a local maximum of (16), it is well known and easy to check (Shapiro 1990) that $f_i'(x_i^*, x_{-i}^*; d_i) \leqslant 0$ for each tangent direction d_i described. This necessary optimality condition is called *B-stationarity* of x_i^*. (Incidentally, B-stationarity has a separate meaning for MPECs; see Scheel and Scholtes 2000.)

We present a relationship between stationary points of MPECs of the form (11) and B-stationary points of their related implicit programs (16). Its proof is technical and in the spirit of the analysis of implicit programs given in Luo et al. (1996, Chapter 4) and is available in the online appendix.

Proposition 6. *Let the above assumptions* (a)–(c) *hold for the general EPEC* $\{(11)\}_{i=1}^N$ *and* $(x^*, y^*, \lambda^*, \mu^*)$ *be a Nash stationary point of this game. If the LICQ holds at* y^* *for the constraints on* y *given by* $H(x^*, y) = 0$ *and* $G(x^*, y) \geqslant 0$, *then for each* i, x_i^* *is B-stationary for the implicit program* (16) *with* $x_{-i} = x_{-i}^*$.

The proposition can be applied to show that Nash stationary points of the bid-a-only EPEC $\{(15)\}_{i=1}^N$ yield B-stationary points of the implicit bid-a-only game $\{(4)\}_{i=1}^N$. The next step is to show that the latter type of stationary point must in fact be a local Nash point. To do this, we use second directional derivatives. If ϕ_i is twice continuously differentiable, then f_i also has second directional derivatives (Ben-Tal and Zowe 1982), including the following:

$$\begin{aligned} &f_i''(x_i, x_{-i}; d_i, d_i) \\ &\quad = \lim_{t \to 0^+} \frac{f_i(x_i + td_i + t^2 d_i, x_{-i}) - f_i(x) - tf_i'(x_i, x_{-i}; d_i)}{t^2}, \end{aligned}$$

which yields a Taylor-type expansion:

$$\begin{aligned} f_i(x_i + td_i, x_{-i}) = f_i(x_i, x_{-i}) + tf_i'(x_i, x_{-i}; d_i) \\ + t^2 f_i''(x_i, x_{-i}; d_i, d_i) + o(t^2). \end{aligned}$$

Suppose that f_i is piecewise quadratic, as it is for the implicit program (4). Then, the Taylor expansion is exact without the $o(t^2)$ error term; hence, x_i is a local maximum of (16) if it is B-stationary and satisfies the second-order condition that $f_i''(x_i, x_{-i}; d_i, d_i) \leqslant 0$ for each feasible direction d_i with $f_i'(x_i, x_{-i}; d_i) = 0$. It can be shown (see the proof of Theorem 3 in the online appendix) that B-stationary points of the implicit program (4) satisfy this second-order condition; hence they are local maximizers, as claimed.

5. Numerical Methods and Examples

The second aspect of our study is completed here by attempting to compute Nash stationary points. This is related to the first aspect, i.e., existence questions, by the discussion in §5.2.3.

We first describe the diagonalization method (§5.1), which is familiar in the computational economics literature from the days of the PIES energy model in Ahn and Hogan (1982) and, more recently, in Cardell et al. (1997). Then, in §5.2 we give computational results for randomly generated electricity market games using both the diagonalization framework, which is implemented via standard nonlinear programming methods, and the CP formulation of §4.3 which requires a complementarity solver.

5.1. Diagonalization Method

Diagonalization is a kind of fixed-point iteration in which players cyclically or in parallel update their strategies while treating other players' strategies as fixed.

Diagonalization method for the game $\{(13)\}_{i=1}^N$:

(i) We are given a starting point $x^{(0)} = (x_1^{(0)}, \ldots, x_N^{(0)})$, iteration counter $k = 0$, maximum number of iterations K, and convergence tolerance $\varepsilon \geqslant 0$.

(ii) We are given the current iterate $x^{(k)}$. For $i = 1$ to N: player i solves problem (13) while holding $x_j = x_j^{(k+1)}$ for $j < i$ and $x_j = x_j^{(k)}$ for $j > i$. Denote the x_i-part of the optimal solution by $x_i^{(k+1)}$.

(iii) *Check stopping condition*: If $\|x_i^{(k+1)} - x_i^{(k)}\| < \varepsilon$ for $i = 1, \ldots, N$, then accept the point and *stop*; else, if $k < K$, then increase k by one, *go to* (ii) and *repeat* the process; else, if $k = K$, *stop* and output "no equilibrium point found."

The stopping condition of the algorithm derives from the observation that if the starting point $x^{(0)}$ is Nash stationary, and the MPEC method used in step (ii) stops at stationary points of the corresponding NLP, then step (ii) will produce $x^{(1)} = x^{(0)}$, a fixed point, and the algorithm will stop (in step (iii)) at this point for any $\varepsilon > 0$. Extending this, suppose that x^* is a local Nash equilibrium at which the MPEC-LICQ holds and, for each i, a second-order sufficient condition holds for player i's problem at x_i^*, given $x_{-i} = x_{-i}^*$. Then, by taking any $x^{(0)}$ near enough to x^*, it is possible for step (ii) to find a local minimizer (and stationary point) $x_i^{(1)}$ such that $\|x_i^{(1)} - x_i^{(0)}\|$ can be made arbitrarily small for all i. In other words, the stopping rule will identify certain local Nash points to within tolerance ε. It may not, however, prevent convergence to a point that is not Nash stationary without further conditions such as those considered in Proposition 7.

In step (ii), we need an algorithm to solve the MPEC (13). There has been a lot of recent progress in numerical methods for solving MPECs, e.g., Anitescu (2005), Fletcher et al. (2006), Hu and Ralph (2004), and references therein, resulting in several different but apparently effective approaches. We could simply apply an off-the-shelf NLP method to an MPEC formulated as an NLP such as (13). An alternative is to use an iterative scheme like the regularization method; see Scholtes (2001). Regularization, like the penalty (Hu and Ralph 2004) and smoothing (Fukushima and Pang 2000) methods, remodels an MPEC as a one-parameter family of better-behaved nonlinear programs, and solves a sequence of the latter while driving the parameter to zero, at which point the NLP becomes equivalent to the MPEC. This leads us to call each diagonalization iteration an outer iteration, and each iteration within step (ii) of diagonalization an inner iteration.

Step (ii) *by regularization method* (inner iteration for ith player in step (ii)).

(ii-1) We are given $x^{(k)}$, z_j for $j < i$, and a sequence of (typically, decreasing) regularization parameters $\rho_0^{(k)}$, $\rho_1^{(k)}, \ldots, \rho_M^{(k)}$, i.e., a fixed number of iterations to execute, where $\rho_m^{(k)}$ bounds the total violation of complementarity in subiteration m. Let $m = 0$, $\xi_i^0 = x_i^{(k)}$.

(ii-2) Given the regularization parameter ρ_m, $x_j = z_j$ for $j < i$ and $x_j = x_j^{(k)}$ for $j > i$, and the starting point $x_i = \xi_i^m$, solve the following nonlinear program:

$$\begin{aligned}
\underset{x_i, y, \lambda, \mu}{\text{maximize}} \quad & f_i(x, y) \\
\text{subject to} \quad & g_i(x_i, y) \geqslant 0, \quad h_i(x_i, y) = 0, \\
& \nabla_y F(x, y) - \nabla_y G(x, y)^T \mu \\
& \qquad + \nabla_y H(x, y)^T \lambda = 0, \\
& H(x, y) = 0, \\
& G(x, y) \geqslant 0, \quad \mu \geqslant 0, \quad G_j(x, y)\mu_j \leqslant \rho_m^{(k)}, \\
& \qquad\qquad j = 1, \ldots, s. \qquad (17)
\end{aligned}$$

If $m < M$, let ξ_i^{m+1} be the x_i part of the solution, let $m = m + 1$, and repeat (ii-2).

Else, let $x_i^{(k+1)}$ be the x_i part of the solution, and stop.

We give a convergence result for diagonalization where, at each iteration, step (ii) is carried out using regularization. Then, we will briefly comment on convergence of diagonalization when step (ii) relies on the direct application of a nonlinear programming solver. Convergence of diagonalization via regularization is essentially a corollary of the convergence analysis of Scholtes (2001), which requires that the "solution" of (17), in (ii-2), is a stationary point that satisfies a second-order necessary condition as described in Scholtes (2001). Assuming that the diagonalization method produces a convergent subsequence, with limit $(x_i^*, y^*, \lambda_i^*, \mu_i^*)$ and limiting multipliers ξ_i^g, ξ_i^h, etc. as laid out in (13), a technical condition called *upper-level strict complementarity* (ULSC) is needed: For each component j of μ_i^* and $G(x^*, y^*)$ such that $(\mu_i^*)_j = G_j(x^*, y^*) = 0$, neither $(\xi_i^G)_j$ nor $(\xi_i^\mu)_j$ is zero.

Proposition 7. *Let M be fixed in* (ii-1)–(ii-2) *for each outer iteration k of the diagonalization method. Assume that $0 < \rho_M^{(k)} \to 0$. Suppose that $x^{(m)} = (x_i^{(m)})_{i=1}^N$ satisfies the following conditions*:

(i) *there exist $Y^{(m)} = (y_i^{(m)})_i$, $\lambda^{(m)} = (\lambda_i^{(m)})_i$, and $\mu^{(m)} = (\mu_i^{(m)})_i$ such that for each $i = 1, \ldots, N$, the tuple $(x_i^{(m)}, y_i^{(m)}, \lambda_i^{(m)}, \mu_i^{(m)})$ is a stationary point satisfying a second-order necessary condition for* (17) *with $x_{-i} = (x_1^{(m-1)}, \ldots, x_{i-1}^{(m-1)}, x_{i+1}^{(m-1)}, \ldots, x_n^{(m-1)})$*;

(ii) *the sequence $(x^{(m)}, Y^{(m)}, \lambda^{(m)}, \mu^{(m)})$ has a limit point $(x^*, Y^*, \lambda^*, \mu^*)$ as $m \to \infty$, where $x^* = (x_i^*)_i$, $Y^* = (y_i^*)_i$, etc.*; *and*

(iii) *for each i, the MPEC-LICQ and ULSC hold at $(x_i^*, y_i^*, \lambda_i^*, \mu_i^*)$ for* (13).

If $y_1^* = \cdots = y_n^*$, then the point x^* is a Nash stationary point of game $\{(9)\}_{i=1}^N$.

We mention that the above diagonalization method is of the Gauss-Seidel type, in that players take turns to update their strategies. An alternative is a Jacobi scheme in which, given the kth iterate $x^{(k)}$, each player i updates its strategy x_i based only on $x^{(k)}$; these updates can be carried out in parallel. A Jacobi scheme and its convergence were discussed in Hu (2003). Further developments, including more details of the convergence analysis supporting the above proposition, and numerical comparisons of the Gauss-Seidel and Jacobi approaches to EPECs that favor the former, appear in Su (2004b).

5.2. Numerical Examples

We implement the three algorithmic methods for (2) with 30 randomly generated problems for each of bid-a-only, bid-b-only, and bid-a-b scenarios on a 3-node network and a 39-node network from Alvarado et al. (2001), respectively. We solve these as EPECs by applying the

complementarity problem formulation of §4.3 and the Gauss-Seidel diagonalization approaches of §5.1.

We have modeled the CP formulation and two versions of the diagonalization methods using GAMS (Brooke et al. 1992) for the bilevel game (2) with three bidding scenarios: bid-a-only, bid-b-only, and bid-a-b, respectively. Our interest is in numerical behavior here; see Hu et al. (2004) for an investigation of the economic implications of equilibria computed for bid-a and bid-b games.

We use PATH (Dirkse and Ferris 1995, Ferris and Munson 1999), version 4.6.04, to solve problems in the CP formulation. If PATH terminates successfully, we then check the second-order sufficient condition for each player's MPEC. In all our successful CP tests below, we were able to establish numerically that a solution found by PATH was indeed a local Nash equilibrium.

By Diag we mean the Gauss-Seidel diagonalization method in which each MPEC in step (ii) of the outer iteration is formulated as an NLP, cf. (11), and solved using CONOPT (Drud 1994). By Diag^{Reg} we mean the Gauss-Seidel scheme using the inner iteration (ii-1)–(ii-2) described above; each inner iteration requires solution of an NLP which is carried out using CONOPT.

We run GAMS, version 21.1, for the problems on a computer with a 1.1 GHz UltraSPARC IIIi Cu processor and 2 GB ram, running the Solaris operating system.

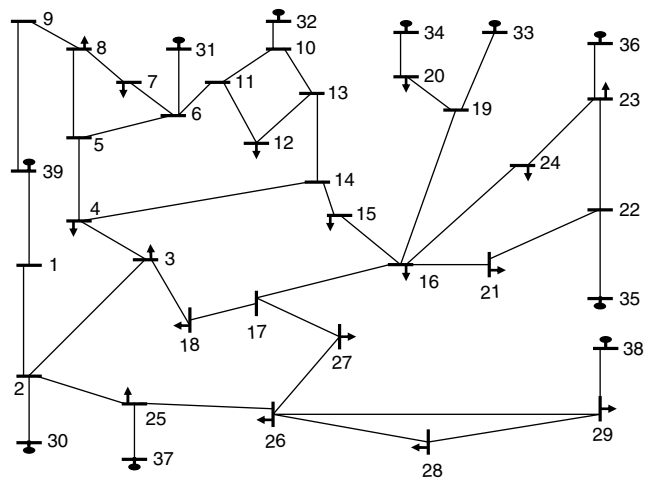
5.2.1. Test Problems. First, regarding the true cost coefficients A_i and B_i, it is generally accepted for such quadratic models (Saadat 1999) that B_i is much smaller than A_i in the units that are commonly employed for electricity market models. Second, we use GAMS's pseudorandom number generator (default seed $= 3{,}141$) to generate 30 sets of uniformly distributed coefficients of generation and utility functions for each of the two networks. For each network, each of the three bidding scenarios is applied to each of the 30 data sets to give a total of 1,800 test problems; see below.

For all test problems, we set the bounds for the a_i and b_i bids as $\underline{A}_i = 0$, $\bar{A}_i = 200.0$, $\underline{B}_i = 0.00001$, and $\bar{B}_i = 6.0$. Ex post checking, for a successfully solved problem, shows that none of these constraints are binding (active) an equilibrium.

All lines in both the 3-node and 39-node networks are assumed to have the same physical characteristics (i.e., the same conductivity), which determine the power distribution factors (see, e.g., Wood and Wollenberg 1996 or Liu and Gross 2004), together with the choice of a reference node.

3-Node Network. The participants are one consumer at node 1, two generators at node 2, and one generator at node 3. However, only the generators optimize their bids, whereas the consumer reveals its true utility function to the ISO. Node 1 is chosen as the reference node, and therefore its distribution factors are zero. The distribution factors of node 2 to lines 1–2, 1–3, 2–3 are $-2/3, -1/3, 1/3$, respectively; and those of node 3 are $-1/3, -2/3, -1/3$, respectively. The transmission limits are $C_{12} = C_{13} = 800.0$ and $C_{23} = 5.0$.

Figure 5. A 39-bus network based on the New England system from Alvarado et al. (2001).



The values of A_1, B_1 for the consumer at node 1 are randomly, uniformly drawn from the intervals $[90.0, 120.0]$ and $[0.2, 0.6]$, respectively; for the generators at node 2 from $[5.0, 10.0]$ and $[0.01, 0.08]$, respectively; and, for the generator at node 3, $[5.0, 20.0]$ and $[0.01, 0.04]$, respectively. The two generators at node 2 are provided with the same cost functions.

39-Node Network. The network from Alvarado et al. (2001) has 39 nodes and 46 links (see Figure 5). There are 17 consumers at the nodes indicated by the withdrawal arrows and 10 generators at the nodes indicated by the bold dots. The intervals given in Tables 1 and 2, from which the 30 sets of random numbers are drawn uniformly, are based on the quadratic cost and utility functions in Alvarado et al. (2001). We have removed the arrows on each link used in Alvarado et al. (2001), which indicate the flow directions for the dispatch given the specific cost and utility functions in their paper.

We impose capacities of 5, 10, and 5 on the three lines 5–6, 14–15, and 16–17, respectively, and the actual numbers of times being congested, out of the number of problems being successfully solved, are given in Table 5 for each bidding scenario.

Table 1. Intervals of randomly generated coefficients of cost functions of the 10 generators.

Node	A	B	Node	A	B	Node	A	B
30	[29, 31]	[0.3, 0.5]	31, 32	[34, 36.5]	[0.2, 0.4]	33	[35, 36.5]	[0.4, 0.6]
34	[35, 36.5]	[0.4, 0.5]	35	[33.5, 35.5]	[0.3, 0.5]	36	[32, 34]	[0.4, 0.6]
37	[35, 37]	[0.2, 0.4]	38	[32.5, 34.5]	[0.2, 0.4]	39	[33.5, 35.5]	[0.2, 0.3]

Table 2. Intervals of randomly generated coefficients of utility functions of the 17 consumers.

Node	A	B	Node	A	B	Node	A	B
3	[52, 72]	[0.3, 0.5]	4	[53.5, 73.5]	[0.4, 0.6]	7	[50, 70]	[0.3, 0.4]
8	[49, 69]	[0.3, 0.5]	12	[50, 70]	[0.4, 0.5]	15	[53, 73]	[0.2, 0.4]
16	[49, 60]	[0.2, 0.4]	18	[55, 75]	[0.3, 0.4]	20	[55, 65]	[0.3, 0.4]
21	[52, 62]	[0.3, 0.5]	23	[53, 73]	[0.4, 0.5]	24	[52, 62]	[0.2, 0.3]
25	[51.5, 61.5]	[0.2, 0.3]	26	[51, 61]	[0.3, 0.4]	27	[52, 72]	[0.3, 0.4]
28	[51.5, 71.5]	[0.3, 0.4]	29	[52, 72]	[0.4, 0.5]			

5.2.2. Details of Parameter Settings and Results of Solver Performance. The true bid parameters for generator cost functions are used as the starting points for all test problems.

Solver settings. For the 39-node network, some settings were changed to improve computation. For example, the preprocessor for PATH solver (Version 4.6.04) was turned off due to execution errors in the processing stage. Also, the maximal number of outer iterations for both Diag and Diag^{Reg} was set to be $K = 400$ except for bid-b-only and bid-a-b games on the 39-node network; in the latter case we used $K = 40$ to force earlier termination on problems for which the code was taking excessive time.

The convergence tolerance for Diag and Diag^{Reg} is $\varepsilon = 10^{-5}$. The norm used to check convergence of the diagonalization schemes at iteration k is the maximum, over all player indices i, of $|a_i^k - a_i^{k-1}| + |b_i^k - b_i^{k-1}|$. In the bid-$a$-only (bid-$b$-only, respectively) situation, this simplifies to the maximum over i of $|a_i^k - a_i^{k-1}|$ ($|b_i^k - b_i^{k-1}|$, respectively).

In the inner iteration process of Diag^{Reg}, given any outer iteration number k, we start with $\rho_0^{(k)} = 10^{-3}$, solve the regularized problem (17), and then divide the regularization parameter by 10. This is repeated five times, for a total of six calls to CONOPT. At the end of the inner iteration process, the maximum violation of the products of pairs of complementary scalars is 10^{-9}.

We also introduce a tolerance for checking Nash stationarity of solutions, $\varepsilon^{\text{stat}} = 10^{-3}$. For the CP formulation, each successful termination of PATH is deemed to be Nash stationary, i.e., the violation of the complementarity system is below 10^{-6} measured in the ∞-norm, which is a PATH default. Checking Nash stationarity for solutions provided by Diag and Diag^{Reg} is performed by applying PATH to the CP formulation using the solution provided as a starting point; if PATH terminates at a point within ∞-distance $\varepsilon^{\text{stat}}$ of the solution provided, then the latter is deemed Nash stationary.

Setting the value of $\varepsilon^{\text{stat}}$ to be orders of magnitude larger than the PATH default stopping tolerance allows for considerable inaccuracy in the solution provided by either of the diagonalization methods. This is a heuristic that allows us to check the claimed success of Diag or Diag^{Reg} given that simple fixed-point iteration generally does not converge faster than linearly, and therefore the termination point has relatively low accuracy. There is, of course, the possibility that Diag or Diag^{Reg} may identify a local Nash point that is not Nash stationary, in which case PATH cannot confirm success.

Performance statistics. We used the following statistics to measure the performance of the solvers or methods used.

 # converged (*# stationary*). For the CP method, this is the number of problems successfully solved by PATH. This number is repeated in brackets because each successful termination of PATH is associated with a Nash stationary point.

For the diagonalization methods, this is the number of problems for which fewer than K outer iterations were needed to achieve convergence. The number in brackets is the number of problems for which PATH verified Nash stationarity of the solution provided (to within tolerance $\varepsilon^{\text{stat}}$ explained above).

Avg. CPU time. The average CPU time is calculated as the total CPU time (given by the GAMS parameter resusd) used by the solver for all convergent problems divided by the number of such problems. For Diag and Diag^{Reg}, the average CPU time includes the time required for all outer and, if applicable, inner iterations (for all convergent problems).

Avg. # outer iter. This is defined only for Diag and Diag^{Reg}. It is the total number of outer iterations k over all convergent problems divided by the number of convergent problems.

Avg. # solver iter. This is the total number of major iterations (given by the GAMS parameter .iterusd) over all convergent problems of either PATH, for the CP formulation, or CONOPT, for the diagonalization formulations, divided by the number of convergent problems.

5.2.3. Numerical Results. We summarise the numerical results below. First, more Nash stationary points are found by PATH, applied to the CP formulation, than by the diagonalization methods. Second, although PATH is the fastest of the three methods on the smaller network, Diag is the clear winner in terms of speed on the larger network. See Tables 3 and 4. Third, we observe that games are much harder to solve on the larger network than on the smaller network when there are transmission constraints, despite using more time in the larger network case. Related to this difficulty for the larger network, we ask if changing bids (e.g., a) often results in a flow pattern (e.g., $q^*(a)$) for which the list of congested links also changes. This conjecture is related to the analysis of §3 and seems supported by Table 5. More detailed discussions follow.

Table 3. Results for three bidding scenarios for the 3-node network with one congested link.

	3-node network with one congested link								
	Bid-*a*-only			Bid-*b*-only			Bid-*a*-*b*		
	CP	Diag	Diag^{Reg}	CP	Diag	Diag^{Reg}	CP	Diag	Diag^{Reg}
# converged (# stationary)	30 (30)	30 (30)	30 (30)	29 (29)	30 (30)	28 (28)	19 (19)	30 (0)	2 (0)
Avg. CPU time	0.14	0.70	5.08	0.37	2.50	15.86	0.56	5.86	8.48
Avg. # outer iter.	N/A	8	8	N/A	26	25	N/A	30	14
Avg. # solver iter.	1,213	96	1,039	2,552	484	2,994	5,775	484	1,642

In bid-*a*-only and bid-*b*-only games on the 3-node network (see Table 3), all methods have similar reliability, converging on most or all problems to solutions that are verified as (approximate) Nash stationary points. All methods also have similar reliability for bid-*a*-only and bid-*b*-only games on the 39-node network (see Table 4), although the number of successful solves and verifications drops to 10–16 out of 30 problem instances. However, significant differences in reliability between PATH and the regularization methods appear when solving bid-*a*-*b* games. There, PATH is by far the most successful in identifying Nash stationary points, although it only solves 35 out of 60 bid-*a*-*b* games. Diag converges in 46 out of 60 cases, but none of the solutions it found could be verified to be Nash stationary. Diag^{Reg} has severe difficulties, converging in only seven out of 60 cases, with Nash stationarity verified in four of these.

We stress that the difficulty experienced by Diag in bid-*a*-*b* games is not in convergence, but in convergence to a Nash stationary point. This leaves open the question of whether or not the diagonalization methods usually find local Nash points. If indeed a local Nash point is not Nash stationary, then, by implication of Proposition 5, the MPEC-LICQ cannot hold there. In any event, difficulties with most computational approaches to bid-*a*-*b* games are to be expected if, as we believe, in contrast to the bid-*a*-only and bid-*b*-only games, the bid-*a*-*b* games have solutions that are not locally unique; see Hu (2003, Chapter 5) for a discussion on nonuniqueness of solutions. Why PATH performs so much better than the diagonalization schemes in finding Nash stationary points of bid-*a*-*b* games is a question for future work.

In terms of speed on convergent problems, PATH is fastest on the 3-node games (Table 3), followed by Diag with Diag^{Reg} the slowest. On the 39-node network (Table 3), Diag is by far the fastest, being roughly an order of magnitude faster than Diag^{Reg} and two orders of magnitude faster than PATH. We conjecture that Diag will be a strong candidate on problems that are perhaps much larger than the 39-node network, a view strengthened by the availability of a range of good NLP solvers.

Finally, we ask whether or not congestion occurred at equilibria. For a discussion on the economic effects of congestion, we refer the readers to Hu et al. (2004). Our interest here relates more to existence; for example, Theorem 3 in §3 suggests that equilibria are likely to exist when the lines are either largely congested over all feasible bids, or largely uncongested. Looking at this another way, if the pattern of congested lines switches as bids change, it may be difficult for a given method to find a Nash stationary point. Behind the results for the 3-node network, we identified around 25 out of 30 tests for each of the three game scenarios, in which congestion occurred in line 2–3 at the solution point identified. The situation for the 39-node network with three capacitated lines is more changeable. Table 5 lists the number of times that capacitated lines are congested for convergent problems. For any game scenario, it can be seen that line 5–6 is usually congested in 50%–75% of these cases; hence, uncongested in 25%–50% of cases, and similarly for line 14–15. Probably, then, for any given 39-node problem, there is a good chance that changing a set of bids will lead to different congestion patterns, which seems to be corroborated by the difficulty in finding equilibria for this network.

Table 4. Results for three bidding scenarios for the 39-node network with three congested links.

	39-node network with three congested links								
	Bid-*a*-only			Bid-*b*-only			Bid-*a*-*b*		
	CP	Diag	Diag^{Reg}	CP	Diag	Diag^{Reg}	CP	Diag	Diag^{Reg}
# converged (# stationary)	16 (16)	16 (16)	16 (16)	10 (10)	13 (13)	13 (13)	19 (19)	16 (0)	5 (4)
Avg. CPU time	119.2	3.40	50.26	195.07	4.07	42.36	172.80	3.60	61.94
Avg. # outer iter.	N/A	5	5	N/A	5	5	N/A	5	5
Avg. # solver iter.	20,878	213	3,587	27,891	245	2,565	22,025	227	4,032

Table 5. Number of times the capacity-limited lines are constrained.

	CP			Diag			Diag^{Reg}		
	Bid-*a*-only	Bid-*b*-only	Bid-*a*-*b*	Bid-*a*-only	Bid-*b*-only	Bid-*a*-*b*	Bid-*a*-only	Bid-*b*-only	Bid-*a*-*b*
# solved	16	10	19	16	13	16	16	13	5
Line 5–6	7	6	13	12	9	12	12	9	5
Line 14–15	7	7	9	9	8	9	9	8	1
Line 16–17	12	7	12	12	10	12	12	10	4

6. Conclusion

Bilevel games and EPECs, unlike classical games with nonempty convex compact strategy spaces, need not admit existence of (pure-strategy) Nash equilibria under convenient general conditions. Nevertheless, we have provided several situations in which the bilevel games considered by Berry et al. (1999) and their EPEC counterparts have Nash equilibria. More generally, we have shown that the stationary conditions of an EPEC can be phrased as a complementarity problem whose solutions characterize Nash stationary points. Computational viability of finding Nash stationary points is shown for a class of randomly generated EPECs in the format of Berry et al. (1999). Of note is the relatively robust behavior of the PATH solver when applied to the complementarity formulation, and the computational promise of diagonalization for larger problems.

7. Electronic Companion

An electronic companion to this paper is available as part of the online version that can be found at http://or.journal.informs.org/.

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